

Commodity Markets Portfolio Optimisation & Risk Analysis

Introduction

This report examines whether naive diversification performs better than mean variance optimisation when investing in commodities. The question matters because mean variance optimization is theoretically attractive, but its practical success depends on whether expected returns and covariances can be estimated accurately enough to generate reliable out of sample portfolio weights. Commodity markets are a useful setting for this test because prices are volatile and shocks to energy, metals or agriculture can affect portfolio performance quickly.

The baseline basket is deliberately energy heavy, including Brent crude oil, Australian coal, US natural gas and European natural gas, because recent commodity instability has been closely linked to energy markets. Russia's invasion of Ukraine sharply worsened existing strains in global energy markets, while the World Bank expected energy prices to rise by around 50% in 2022 and non energy prices by about 20% (IEA, 2022; World Bank, 2022). The core comparison in this report is between a naive 1/N portfolio and an MVO portfolio with varying weights. As an extension, a global minimum variance portfolio is included to help explain whether any MVO underperformance is due to optimisation generally, or specifically to noisy expected return estimates.

Literature and theoretical framework

Modern portfolio theory suggests that investors should consider both expected return and risk when choosing portfolios. Markowitz's mean variance framework shows that diversification can reduce portfolio variance when assets are not perfectly correlated, and that investors are able to choose an efficient portfolio by balancing expected return against variance (Markowitz, 1952). From this perspective, MVO should perform better than naive equal weighting because it uses information on expected returns, variances and covariances rather than investing the same amount in every asset.

The practical difficulty is that these inputs are estimated rather than known. Small errors in expected returns or covariances can produce large changes in optimised weights. Michaud (1989) argues that mean variance optimisation can become an "error maximiser" because it places large weights on assets with favourable estimated inputs, even when those estimates partly reflect noise. This is directly relevant to the set question because MVO is allowed to

choose varying weights, while $1/N$ avoids this estimation problem by allocating mechanically.

This explains why $1/N$ may be difficult to beat in practice. DeMiguel, Garlappi and Uppal (2009) show that many optimised strategies struggle to outperform $1/N$ out of sample, because the theoretical gains from optimisation can be offset by estimation error. In this report, that insight provides the central hypothesis, if commodity return estimates are unstable, naive diversification may outperform MVO despite being much simpler.

GMV is included as a further measure within this report because it uses the covariance matrix but does not require expected return estimates. Therefore, if both $1/N$ and GMV outperform MVO, this would suggest that MVO's weakness is linked especially to noisy expected return estimation rather than to risk based optimisation itself. Commodity markets make this comparison particularly relevant because they are exposed to supply disruptions, storage constraints, geopolitical shocks and changing demand conditions. Financialisation may also have increased co-movement across some commodity markets, meaning correlations are not fixed over time (Tang and Xiong, 2012). Recent energy instability strengthens this concern, as the IMF reports that commodity prices rose by 19.1% between February and August 2022, led by energy prices, with natural gas rising by 129.2% as Russia cut gas supplies to Europe (IMF, 2022). This makes commodities a challenging environment for MVO, because both expected returns and risk relationships can shift quickly.

Data and empirical design

The empirical analysis uses monthly commodity price data from the World Bank Pink Sheet. The baseline basket contains 11 commodities: Brent crude oil, Australian coal, US natural gas, European natural gas, soybeans, maize, US hard red winter wheat, aluminium, copper, gold and silver. This selection deliberately gives energy a large role because oil, gas and coal have been central to recent commodity instability. At the same time, the inclusion of agriculture, industrial metals and precious metals keeps the portfolio broad enough to test diversification across different commodity groups (World Bank, 2026).

Prices are converted into monthly log returns, as portfolio performance depends on price changes over price levels. The report compares three allocation rules. The $1/N$ portfolio invests equally across all commodities. GMV chooses weights using the covariance matrix to reduce portfolio variance and MVO uses both estimated mean returns and covariances,

making it the most information intensive strategy and therefore the most exposed to estimation error.

The main specification uses a 60-month rolling formation window with quarterly rebalancing. At each rebalance date, the model estimates weights from the previous five years of returns and then applies those weights out of sample. The baseline version only allows positive weights, requires all weights to sum to one, and caps any individual commodity at 40%. These restrictions keep the portfolios economically realistic while still allowing the optimised strategies to move away from equal weighting. Although the baseline specification uses realistic portfolio restrictions, an unconstrained version of MVO is also tested to examine whether these restrictions limit performance or prevent the optimiser from taking extreme positions.

Portfolio outcomes are assessed using mean return, standard deviation, Sharpe ratio, semi-variance, maximum drawdown, historical value at risk and expected shortfall. To explain the results rather than only rank the portfolios, the analysis also tracks turnover, concentration, maximum single commodity weight, cap binding months and weight volatility. Additional checks then vary the rolling window and rebalancing frequency, expand the commodity basket, compare capped and unconstrained optimisation, split the sample around the post 2022 energy shock, and test a smaller tradeable style proxy subset.

Main empirical results

Table 1:

Portfolio	Mean return	Std. dev.	Sharpe ratio	Semi-variance	Hist. <u>VaR</u> 5%	Expected shortfall 5%	Max drawdown
1/N	0.0029	0.0345	0.0851	0.0011	0.0512	0.0744	-0.6100
GMV	0.0036	0.0276	0.1297	0.0007	0.0368	0.0568	-0.5260
MVO	-0.0003	0.0453	-0.0074	0.0020	0.0704	0.1055	-0.8888

The main results answer the question clearly, the 1/N portfolio performs better than MVO in the baseline commodity portfolio. Table 1 shows that 1/N produces a positive mean monthly return of 0.0029 and a Sharpe ratio of 0.0851, while MVO produces a slightly negative mean return of -0.0003 and a negative Sharpe ratio of -0.0074. This means that the more complex MVO strategy does not improve performance relative to simple naive diversification. Instead, it performs worse both in average return terms and on a risk adjusted basis.

The downside measures strengthen this conclusion. MVO has the highest standard deviation, at 0.0453, compared with 0.0345 for 1/N. It also has the worst semi-variance, the largest 5% value at risk and the largest expected shortfall. Its maximum drawdown is especially severe, at -0.8888, compared with -0.6100 for 1/N. Therefore, MVO does not simply fail to earn a higher return; it also exposes the portfolio to much larger downside losses. This supports the view that the theoretical gains from mean variance optimisation are being outweighed by estimation error in practice.

GMV performs best overall, with the highest mean return of 0.0036, the lowest standard deviation of 0.0276 and the highest Sharpe ratio of 0.1297. This finding adds nuance to the answer. The evidence does not imply that all optimisation fails. Instead, it suggests that risk-based optimisation can be useful when it avoids direct reliance on expected return estimates. GMV uses the covariance matrix but does not forecast mean returns, while MVO uses both. The fact that GMV performs much better than MVO indicates that the expected return component of MVO is likely the main source of fragility.

Figure 1:

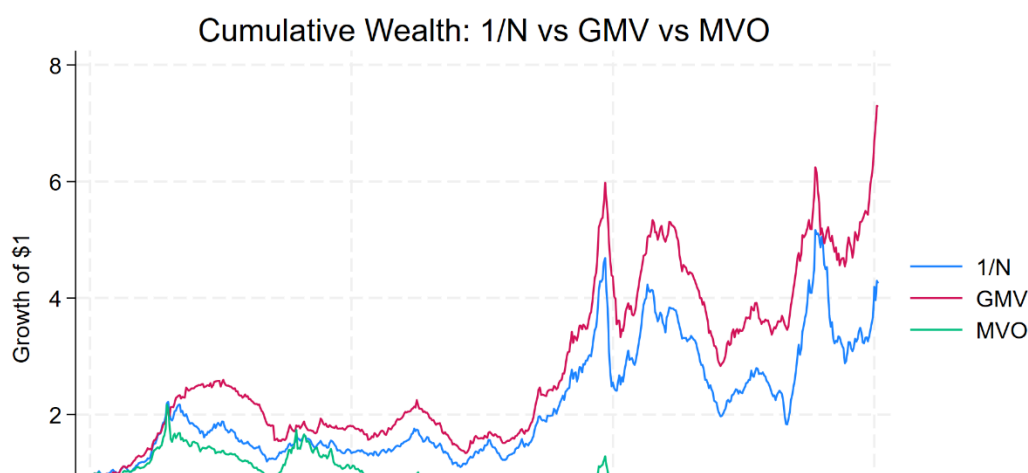


Figure 1 supports the same pattern visually. GMV generates the strongest cumulative wealth path, while 1/N remains clearly above MVO for much of the out of sample period. MVO performs particularly poorly after the late 2000s and does not recover relative to the other two portfolios. This matters because the rolling design tests portfolio rules outside the estimation window. The poor cumulative path of MVO suggests that weights which look attractive using past returns do not translate into stronger future performance.

Table 2:

Portfolio	Avg turnover	Avg concentration	Avg max weight	Cap months	Avg weight SD
1/N	0.0000	0.0909	0.0909	0	0.0000
GMV	0.0261	0.2256	0.3587	313	0.0808
MVO	0.0768	0.3142	0.3958	565	0.1355

Table 2 explains why MVO performs worse than 1/N. The equal weighted portfolio has zero turnover, a concentration score of 0.0909 and a maximum weight of 0.0909, reflecting its fixed and fully diversified allocation across 11 commodities. MVO is also far more unstable, its average turnover is 0.0768, almost three times GMV's 0.0261, and its concentration is higher at 0.3142, compared with 0.2256 for GMV.

The cap binding results are particularly important. MVO's average maximum weight is 0.3958, almost exactly at the 40% cap, and it reaches this cap in 565 months. GMV reaches the cap in 313 months and has a lower average maximum weight of 0.3587. MVO also has greater weight volatility, with an average weight standard deviation of 0.1355, compared with

0.0808 for GMV. This shows that MVO repeatedly shifts heavily into whichever commodity appears attractive from recent rolling estimates.

These results support the estimation error argument. MVO reacts strongly to noisy expected return estimates, becomes concentrated, and changes weights more aggressively, but this does not improve out of sample performance. Instead, 1/N beats MVO because it avoids this estimation risk entirely. GMV's stronger performance adds nuance in that optimisation can help when it focuses on risk control, but full mean return optimisation appears too fragile in this commodity setting.

Sensitivity checks and further analysis

Table 3:

Specification	1/N Sharpe	GMV Sharpe	MVO Sharpe	Key interpretation
60-month quarterly rebalancing	0.0560	0.0989	-0.0330	1/N beats MVO; GMV performs strongest
36-month quarterly rebalancing	0.0815	0.1302	-0.0601	MVO weakens under a shorter window
84-month quarterly rebalancing	0.0679	0.0928	-0.0475	Longer estimation does not rescue MVO
60-month annual rebalancing	0.0562	0.0898	-0.0461	Less frequent rebalancing leaves MVO weakest
14 asset basket	0.0916	0.1437	-0.0413	Wider diversification strengthens 1/N and GMV, not MVO
Post 2022 stress period	0.0395	0.2025	0.0322	MVO improves, but remains below 1/N and GMV
Tradeable style proxy subset	0.0699	0.1207	0.0076	The ranking broadly survives in a more investable subset

Table 3 reports Sharpe ratios from additional portfolio checks. The 14 asset basket adds WTI, Japan LNG and platinum to the baseline portfolio. The tradeable style proxy subset uses WTI, US natural gas, gold, copper and wheat. The post 2022 period captures the recent energy stress environment discussed earlier in the report.

Table 3 tests whether the main 1/N versus MVO result depends on one particular modelling choice. Across the 36-month, 60-month and 84-month rolling windows, and under annual rebalancing, 1/N keeps a positive Sharpe ratio while MVO remains negative which matters because each specification changes the information available to the optimiser. If MVO were extracting reliable return signals, its performance should be more stable across these settings. Instead, the results support DeMiguel, Garlappi and Uppal's (2009) argument that estimation error can remove the theoretical advantage of optimisation.

The wider basket check strengthens this interpretation. When the commodity universe expands from 11 to 14 assets, the 1/N Sharpe ratio rises to 0.0916, while MVO falls to -0.0413. A larger basket could have helped MVO by giving it more allocation opportunities however, it appears to give the optimiser more noisy expected return estimates to overreact to, consistent with Michaud's (1989) "error maximiser" critique.

The constraint test gives the clearest evidence of this instability. When restrictions are removed, unconstrained MVO produces average gross exposure of 11.1714, an average maximum weight of 3.4051 and an average minimum weight of -2.4236. These are not realistic commodity weights. They show that MVO becomes extreme when allowed to act freely on estimated returns, which supports the use of capped weights in the main specification.

Finally, the post 2022 stress check links the results to the energy-market context introduced earlier. MVO improves during this period, but remains more downside exposed, while the smaller tradeable style subset again shows 1/N outperforming MVO. Overall, the additional checks reinforce the central answer: naive diversification performs better than MVO in this commodity setting, while GMV's stronger results suggest that risk focused allocation is more reliable than full mean-return optimisation.

Conclusion and implications

This report finds that naive diversification performs better than mean variance optimisation in the commodity portfolio analysed. In the baseline results, 1/N records a positive Sharpe ratio of 0.0851, while MVO records a negative Sharpe ratio of -0.0074. MVO also performs worse on downside risk, with a maximum drawdown of -0.8888, compared with -0.6100 for 1/N. This suggests that MVO does not deliver a better risk return trade off in this commodity setting.

The mechanism evidence explains this result. MVO has higher turnover, greater concentration and reaches the 40% cap in 565 months, compared with 313 months for GMV. This indicates that MVO repeatedly shifts heavily into commodities that look attractive from recent rolling estimates, but those signals do not hold out of sample. The further checks support the same conclusion, as 1/N continues to outperform MVO across alternative windows, a wider 14 asset basket and a smaller tradeable style subset.

The inclusion of GMV adds an important qualification. GMV achieves the strongest baseline Sharpe ratio of 0.1297 and the least severe drawdown of -0.5260, suggesting that optimisation itself is not the main problem. Instead, the weakness lies in MVO's reliance on noisy expected return estimates. However, GMV is not risk free, it is more concentrated than 1/N and still depends on estimated covariance relationships, which may shift during commodity stress. The analysis also relies mainly on World Bank spot or reference prices rather than direct futures returns. Overall, the evidence supports a nuanced answer, simple diversification beats MVO, while risk focused optimisation can add value when it avoids unstable return forecasts.

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